



# **Energy Information Administration Time Series Report:** **Predicting Weekly U.S. Product Supplied of Petroleum** **Products**

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## Executive Summary

This report analyzes the weekly U.S. Total Products Supplied of Petroleum Products, a key indicator of overall energy demand. The report uses data from November 1990 through October 2025 provided by the U.S. Energy Information Administration (EIA). The objective of the report is to identify long term trends, seasonal patterns, and forecast short term petroleum supply of the United States.

Since 1990, the total products supplied have generally increased, which reflects economic expansion, increases in transportation activity, and industrial growth. There were two major disruptions to the U.S. petroleum market which were clearly visible in the data. The first being the 2008 global financial crisis, which caused a significant drop in supply. The second noticeable disruption was the 2020 COVID-19 pandemic, which produced the steepest supply decline in the dataset. Since mid 2021 the U.S. petroleum supply has stabilized at around 19 million barrels per day, which indicates a steady and mature market in recent years.

The goal of the project was to apply two different time series models on the dataset, understand what transformation would suit the data, and then compare the performance of the chosen models. Following differencing to stabilize the mean, the two chosen models were as follows. The non-seasonal ARIMA(5,1,1) captured short-term patterns effectively, while the seasonal SARIMA(0,1,1)(0,1,1) 52 better represented annual cyclicity. Both models produced reasonable forecasts, but the SARIMA model provided a more comprehensive picture of underlying seasonality and long-run equilibrium behavior in U.S. petroleum demand.

## Introduction

The Energy Information Administration's (EIA) *Total Products Supplied* series represents the volume (thousands of barrels per day) of refined petroleum products consumed domestically each week. It is a fundamental measure of U.S. economic and transportation activity. The metric WRPUPUS2 in our data is the aggregated barrel supplied by all fuel types, including regular gasoline, distillates, fuel oils, jet fuel and others. Total products supplied provides a comprehensive measure of petroleum demand. Figure 1 shows the proportions of petroleum products that make up the total petroleum products supplied over time. One thing to note would be the decline in residual fuel oil and slight increase in gasoline over time. Understanding this series helps policymakers, refiners, and economists anticipate shifts in energy consumption and adjust production accordingly. This report focuses on characterizing the long-term behavior of total petroleum demand and developing simple, time series models that can produce short-term forecasts.

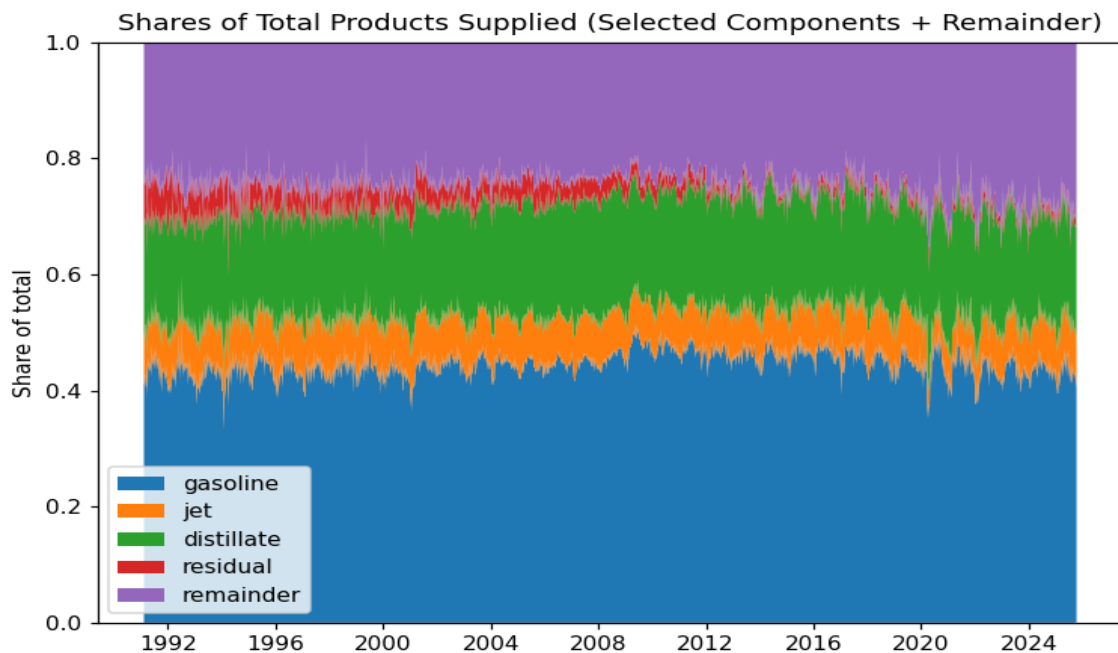


Figure 1 – Share Plot

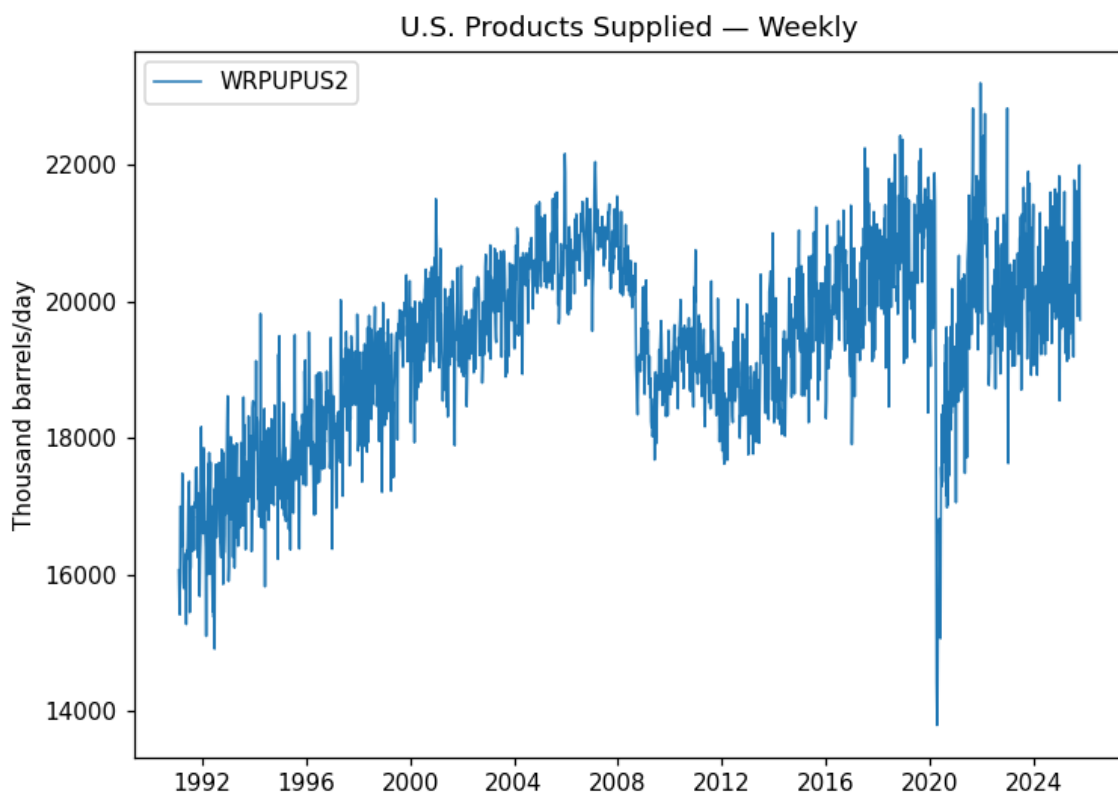
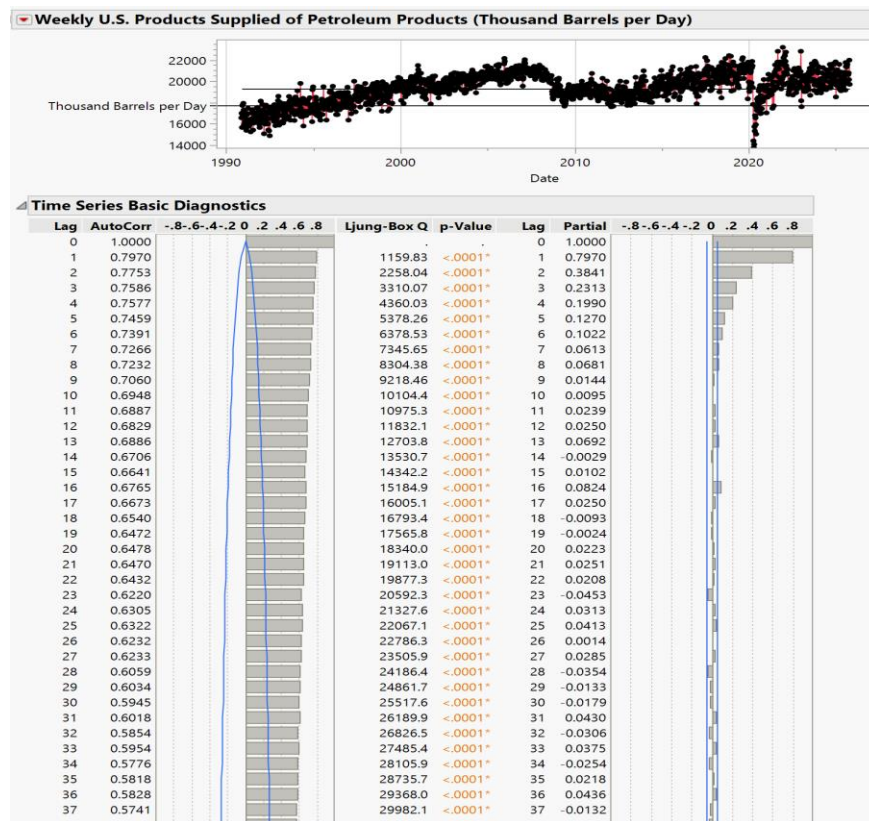


Figure 2 – Time Series Plot

This figure shows the weekly total petroleum products supplied in the U.S. from 1990 through 2025. The graph rises steadily through the 1990s and early 2000s, dips during the 2008–09 financial crisis and drops sharply in 2020 during COVID. Since then, demand has leveled off at nearly 19 to 20 million barrels per day.

## Model Building

Beginning the modeling analysis in JMP, a time series model was created using total Weekly Products Supplied of Petroleum Products (WRPUPUS2) and the provided dates. The initial plot shown below aligns with Figure 1 produced in Python and shows a clear upward trend. When considering stationarity, the presence of trends, unstable variance, or correlated errors can indicate the need for differencing. While the model's variance shows no obvious patterns, the upward trending mean, and slowly decaying autocorrelations indicate that the series is non-stationary and requires differencing.



After differencing the data, as shown below, the mean now exhibits a more constant pattern, and the autocorrelation lags decay quickly. The differenced series looks much more stable and stationary.

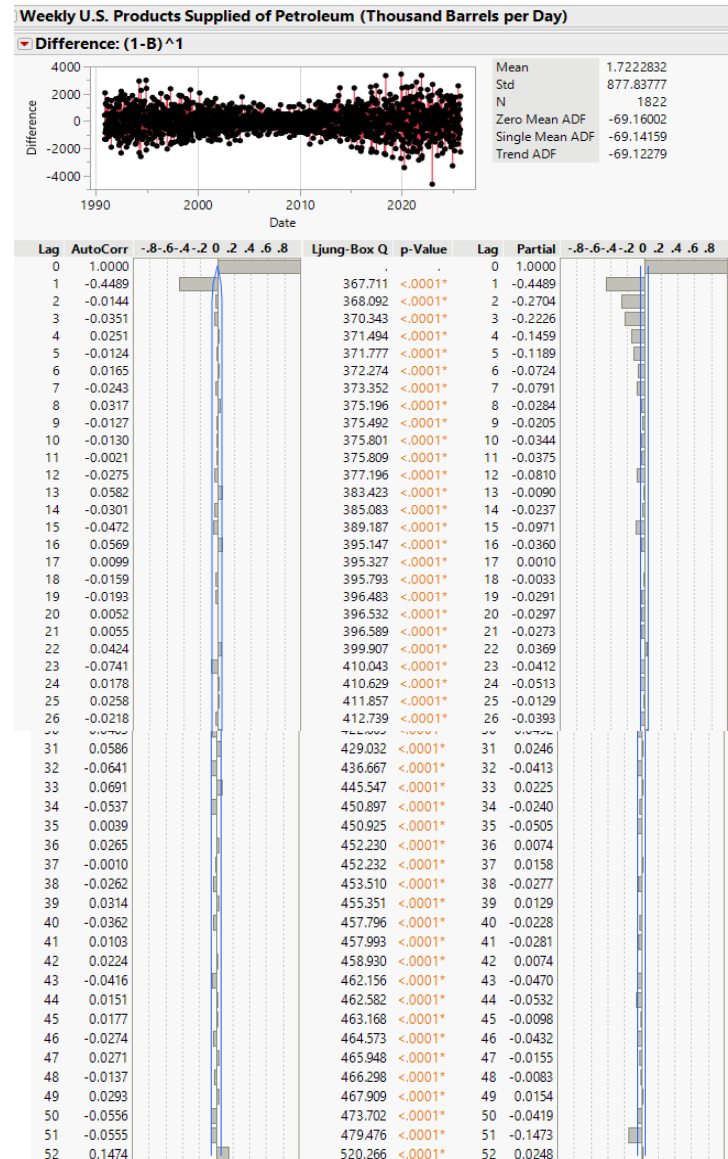
The next steps included examining the differenced ACF and PACF for possible model orders and seasonality. In regard to seasonality, the autocorrelations did not exhibit strong seasonal patterns. As the data is weekly, we would expect to see strong seasonal patterns in lags that are multiples of 7. This did not appear. However, there was 1 significant lag at 52. This could point to some annual seasonality in our data, so we will approach modeling both with and without seasonal influences.

### Non-Seasonal Model

Starting with a non-seasonal model, significant lags could be seen on the ACF at lag 1 and the PACF at lags 1, 2, 3, 4, and 5. This could point to a Moving Average(1) and an Autoregressive(5). To best encompass all possible non-seasonal models, an ARIMA Model group was performed, which allows the ability to evaluate multiple combinations of these model orders.

In the ARIMA Model Group, as seen below, we set both Autoregressive(AR) and Moving Average(MA) to 5 to capture as many prospective models as possible. This led to 36 models.

| Report                   | Graph                    | Model          | DF   | Variance  | AIC       | SBC       |
|--------------------------|--------------------------|----------------|------|-----------|-----------|-----------|
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(1, 1, 2) | 1818 | 508853.6  | 29116.733 | 29138.764 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(1, 1, 1) | 1819 | 512660.71 | 29129.017 | 29145.540 |
| <input type="checkbox"/> | <input type="checkbox"/> | IMA(1, 2)      | 1819 | 512913.67 | 29129.908 | 29146.431 |
| <input type="checkbox"/> | <input type="checkbox"/> | IMA(1, 1)      | 1820 | 515006.48 | 29136.290 | 29147.305 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(2, 1, 1) | 1818 | 512284.57 | 29128.706 | 29150.737 |
| <input type="checkbox"/> | <input type="checkbox"/> | IMA(1, 3)      | 1818 | 512424.41 | 29129.191 | 29151.221 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(2, 1, 3) | 1816 | 509398.28 | 29120.676 | 29153.723 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(4, 1, 2) | 1815 | 508557    | 29118.669 | 29157.223 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(1, 1, 5) | 1815 | 508570.9  | 29118.717 | 29157.271 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(2, 1, 2) | 1817 | 512494.81 | 29130.437 | 29157.975 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(3, 1, 1) | 1817 | 512497.2  | 29130.472 | 29158.010 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(1, 1, 3) | 1817 | 512642.97 | 29130.954 | 29158.493 |
| <input type="checkbox"/> | <input type="checkbox"/> | IMA(1, 4)      | 1817 | 512701.91 | 29131.176 | 29158.715 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(4, 1, 1) | 1816 | 510948.82 | 29126.047 | 29159.093 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(3, 1, 3) | 1815 | 508995.42 | 29121.032 | 29159.586 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(3, 1, 4) | 1814 | 507878.37 | 29117.315 | 29161.376 |
| <input type="checkbox"/> | <input type="checkbox"/> | ARIMA(5, 1, 1) | 1815 | 510156.52 | 29124.276 | 29162.829 |
| <input type="checkbox"/> | <input type="checkbox"/> | IMA(1, 5)      | 1816 | 512254.79 | 29130.611 | 29163.657 |



ARIMA Model Group

Specify ARIMA Model

| ARIMA                   |     | Seasonal ARIMA          |     |
|-------------------------|-----|-------------------------|-----|
| p, Autoregressive Order | 0 5 | P, Autoregressive Order | 0 0 |
| d, Differencing Order   | 1 1 | D, Differencing Order   | 0 0 |
| q, Moving Average Order | 0 5 | Q, Moving Average Order | 0 0 |
|                         |     | Observations per Period | 7 7 |

Prediction Interval: 0.95

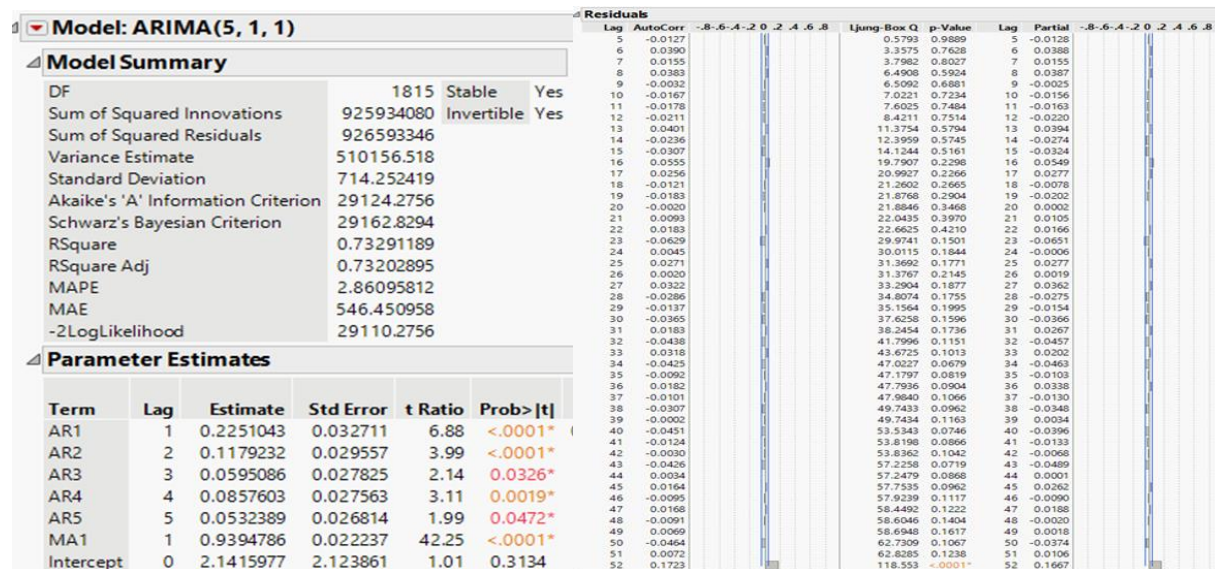
☒ Intercept

☒ Constrain fit

Total Number of Models: 36

Estimate Cancel Help

All 36 models were reviewed, and the top performing non-seasonal model was an ARIMA(5,1,1). This model was ranked 17<sup>th</sup> for the SBC(29,162.83), all parameters were significant, and residuals were mostly consistent with white noise, with a few violations(lag 52).



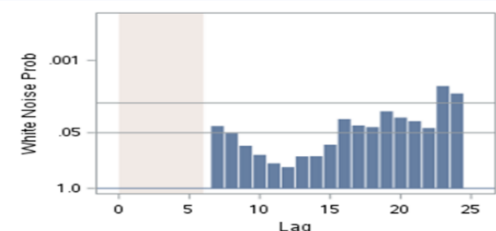
In an attempt to improve the model, outliers and level shifts were also found and added. 15 Additive Outliers(AO) and Level Shifts(LS) were found in total.

| Outlier Detection Summary |       | Constant Estimate   | 2.316272 |
|---------------------------|-------|---------------------|----------|
| Maximum number searched   | 30    | Variance Estimate   | 419628.5 |
| Number found              | 15    | Std Error Estimate  | 647.7874 |
| Significance used         | 0.001 | AIC                 | 28784.17 |
|                           |       | SBC                 | 28905.34 |
|                           |       | Number of Residuals | 1822     |

| Maximum Likelihood Estimation |          |                |         |                |     |          |
|-------------------------------|----------|----------------|---------|----------------|-----|----------|
| Parameter                     | Estimate | Standard Error | t Value | Approx Pr >  t | Lag | Variable |
| MU                            | 4.20816  | 0.77688        | 5.42    | <.0001         | 0   | WRPUPUS2 |
| MA1,1                         | 0.97418  | 0.0071975      | 135.35  | <.0001         | 1   | WRPUPUS2 |
| AR1,1                         | 0.16284  | 0.02482        | 6.56    | <.0001         | 1   | WRPUPUS2 |
| AR1,2                         | 0.08303  | 0.02471        | 3.36    | 0.0008         | 2   | WRPUPUS2 |
| AR1,3                         | 0.04656  | 0.02466        | 1.89    | 0.0590         | 3   | WRPUPUS2 |
| AR1,4                         | 0.08834  | 0.02459        | 3.59    | 0.0003         | 4   | WRPUPUS2 |
| AR1,5                         | 0.06882  | 0.02448        | 2.81    | 0.0049         | 5   | WRPUPUS2 |
| NUM1                          | -4621.9  | 579.60044      | -7.97   | <.0001         | 0   | LS1535   |
| NUM2                          | 2358.9   | 371.29978      | 6.35    | <.0001         | 0   | LS1544   |
| NUM3                          | 3023.5   | 627.59933      | 4.82    | <.0001         | 0   | AO1677   |
| NUM4                          | 1893.9   | 268.58277      | 7.05    | <.0001         | 0   | LS1597   |
| NUM5                          | -2279.3  | 501.75756      | -4.54   | <.0001         | 0   | LS1533   |
| NUM6                          | -2532.1  | 628.19982      | -4.04   | <.0001         | 0   | AO1518   |
| NUM7                          | 2547.0   | 628.88336      | 4.07    | <.0001         | 0   | AO178    |
| NUM8                          | 2420.8   | 628.96081      | 3.85    | 0.0001         | 0   | AO1623   |
| NUM9                          | -2274.8  | 625.67121      | -3.64   | 0.0003         | 0   | AO1439   |
| NUM10                         | 2066.3   | 474.58765      | 4.35    | <.0001         | 0   | LS1538   |
| NUM11                         | -1306.8  | 261.12261      | -5.00   | <.0001         | 0   | LS1638   |
| NUM12                         | -2063.7  | 626.47491      | -3.29   | 0.0010         | 0   | AO1574   |
| NUM13                         | -1775.1  | 267.24974      | -6.64   | <.0001         | 0   | LS932    |
| NUM14                         | -2037.7  | 626.77337      | -3.25   | 0.0011         | 0   | AO1782   |
| NUM15                         | -2122.4  | 629.04455      | -3.37   | 0.0007         | 0   | AO1679   |

When adding the outliers and level shifts into the model, it was found that all AOs/LSs were significant. Meaning they all helped in explaining variation in the data. However, even though this process improved the SBC from 29,162.83 to 28,905.34, this also led to one of the parameters (AR1,3) becoming insignificant. The residuals had more violations as well. Adding outliers can improve the forecasting accuracy as it helps maintain relevant shock events. So, the model forecasts with/without outliers was compared to the data's actual values.



| Dates        | Actual Product Supplied | Model 1 without AO/LS Prediction | Model 2 with AO/LS Prediction | Model 1 Error % | Model 2 Error % |
|--------------|-------------------------|----------------------------------|-------------------------------|-----------------|-----------------|
| Jun 06, 2025 | 19762                   | 19892.1737                       | 20066.9304                    | -1%             | -2%             |
| Jun 13, 2025 | 20391                   | 19893.9668                       | 20072.4816                    | 2%              | 2%              |
| Jun 20, 2025 | 20513                   | 20101.2449                       | 20230.8752                    | 2%              | 1%              |
| Jun 27, 2025 | 20487                   | 20192.7059                       | 20274.9041                    | 1%              | 1%              |
| Jul 04, 2025 | 20863                   | 20239.5096                       | 20289.4134                    | 3%              | 3%              |
| Jul 11, 2025 | 19184                   | 20433.4751                       | 20442.9578                    | -7%             | -7%             |
| Jul 18, 2025 | 21770                   | 20067.6665                       | 20223.4327                    | 8%              | 7%              |
| Jul 25, 2025 | 21387                   | 20582.4098                       | 20570.9838                    | 4%              | 4%              |
| Aug 01, 2025 | 20122                   | 20781.7556                       | 20699.9677                    | -3%             | -3%             |
| Aug 08, 2025 | 21357                   | 20442.8133                       | 20447.516                     | 4%              | 4%              |
| Aug 15, 2025 | 21506                   | 20737.5161                       | 20664.4706                    | 4%              | 4%              |
| Aug 22, 2025 | 21614                   | 20993.7803                       | 20900.5452                    | 3%              | 3%              |
| Aug 29, 2025 | 20652                   | 21018.7515                       | 20870.6295                    | -2%             | -1%             |
| Sep 05, 2025 | 19781                   | 20841.1367                       | 20748.6043                    | -5%             | -5%             |
| Sep 12, 2025 | 20637                   | 20553.4419                       | 20607.4126                    | 0%              | 0%              |
| Sep 19, 2025 | 20793                   | 20609.3773                       | 20652.5737                    | 1%              | 1%              |
| Sep 26, 2025 | 20166                   | 20628.9132                       | 20636.886                     | -2%             | -2%             |
| Oct 03, 2025 | 21990                   | 20404.128                        | 20434.6024                    | 7%              | 7%              |
| Oct 10, 2025 | 19726                   | 20874.0328                       | 20744.9831                    | -6%             | -5%             |

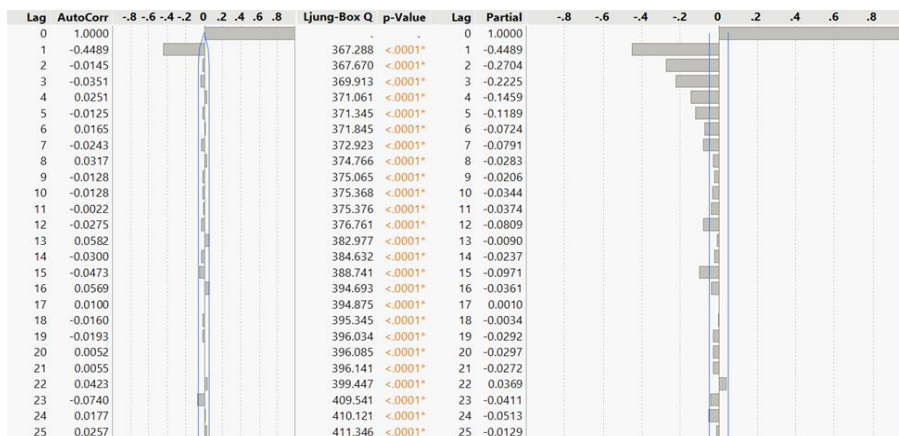
Model forecasts were compared over the last seven months for both versions. The total error differed by only 1%, indicating that adding additional outliers did not meaningfully improve forecasting accuracy. Because the residuals also moved further from white noise after including these outliers as well as the model was no longer parsimonious, we chose to use the original ARIMA(5,1,1) as the non-seasonal model.

## Seasonal Model

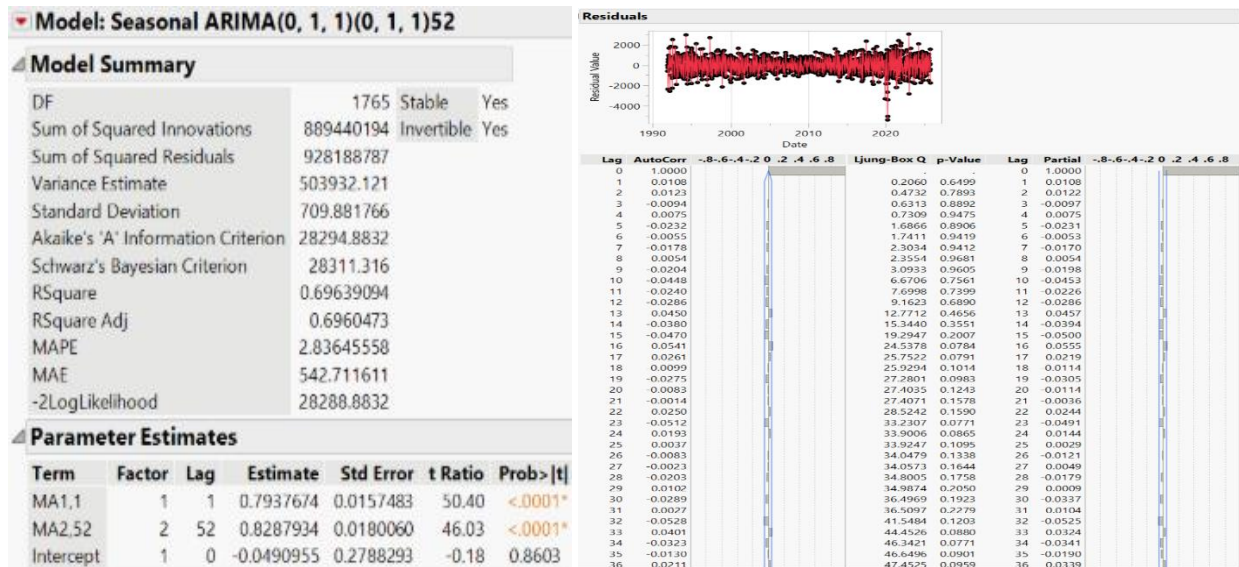
When differencing the data, significant lags were also found on lag 52 for the ACF and 51 for the PACF.



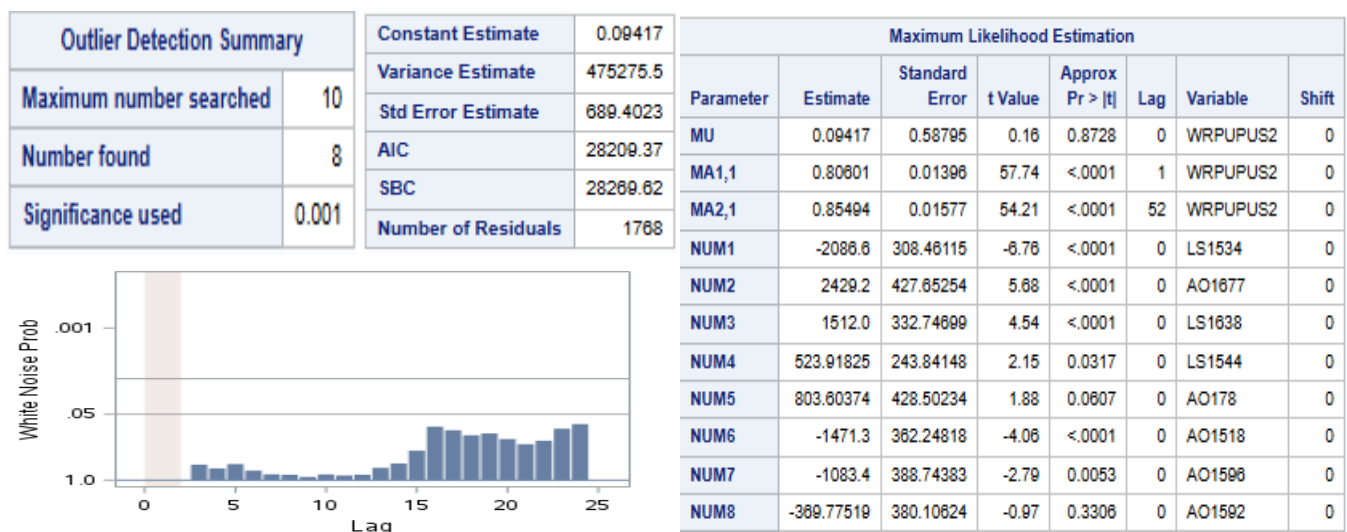
This indicated that the time series may have some annual seasonality. It made sense intuitively for the EIA data, because during different times of the year, petroleum products will have differing demands. In the search for a seasonal model, models with non-seasonal and seasonal differencing were considered. In the image below, the autocorrelations display a non-seasonal and seasonal differenced time series.



The ACF had a significant lag of 1, and the PACF had significant lags of 1 through 5. Many SARIMA models were examined through the ARIMA Model Group and it was found that SARIMA(0,1,1)(0,1,1)<sub>52</sub> performed the best. It had the lowest AIC and SBC, while being valid and parsimonious.



Again, in an attempt to improve the model, outliers and level shifts were also found and added. 8 Additive Outliers(AO) and Level Shifts(LS) were found in total.



Adding the outliers and level shifts to this model had somewhat differing results from the non-seasonal model. Here, not all outliers were significant, but the model parameters and residuals remained valid and parsimonious. The SBC also improved from 28311.316 to 28269.62. Like the last model, the forecasting was tested with and without outliers.

| Dates        | Actual Product Supplied | Model 1 without AO/LS Predictions | Model 2 with AO/LS Predictions | Model 1 Error % | Model 2 Error % |
|--------------|-------------------------|-----------------------------------|--------------------------------|-----------------|-----------------|
| Aug 08, 2025 | 21357                   | 20904.0753                        | 20906.5208                     | 2%              | 2%              |
| Aug 15, 2025 | 21506                   | 21053.3743                        | 21064.0785                     | 2%              | 2%              |
| Aug 22, 2025 | 21614                   | 21156.3281                        | 21100.3911                     | 2%              | 2%              |
| Aug 29, 2025 | 20652                   | 20628.3535                        | 20639.4385                     | 0%              | 0%              |
| Sep 05, 2025 | 19781                   | 20333.6972                        | 20381.337                      | -3%             | -3%             |
| Sep 12, 2025 | 20637                   | 20473.4624                        | 20496.7605                     | 1%              | 1%              |
| Sep 19, 2025 | 20793                   | 20677.5327                        | 20659.4062                     | 1%              | 1%              |
| Sep 26, 2025 | 20166                   | 20562.4433                        | 20600.2663                     | -2%             | -2%             |
| Oct 03, 2025 | 21990                   | 20408.9633                        | 20420.1727                     | 7%              | 7%              |
| Oct 10, 2025 | 19726                   | 21475.8629                        | 21431.4353                     | -9%             | -9%             |

The seasonal models were compared in the same process as the non-seasonal models, where the last 7 months of the forecast was calculated. Again, the performance of both models was extremely close. When adding up the absolute percentage of the errors over the time frame, the model without outliers actually performed 2% better. So, the original SARIMA(0,1,1)(0,1,1)<sub>52</sub> model was chosen.

### Forecasting Future Values

When the EIA data was selected, the timeframe went through October 10, 2025. However, since then, EIA has continued to update the Petroleum Product Supplied on a weekly basis. This allows us to forecast beyond our original dataset and still compare the predictions to the newly posted values. The online dataset now runs through November 7, 2025, so we will generate forecasts and compare them accordingly.

| Future Dates | Actual Values | ARIMA(5,1,1) Predictions | SARIMA(0,1,1)(0,1,1) <sub>52</sub> Predictions | Model 1 Error % | Model 2 Error % |
|--------------|---------------|--------------------------|------------------------------------------------|-----------------|-----------------|
| Oct 17, 2025 | 20014         | 20532.6805               | 20622.4509                                     | -2.59%          | -3.04%          |
| Oct 24, 2025 | 21281         | 20511.3436               | 20774.9153                                     | 3.62%           | 2.38%           |
| Oct 31, 2025 | 20356         | 20590.9573               | 20910.0655                                     | -1.15%          | -2.72%          |
| Nov 07, 2025 | 20770         | 20558.3136               | 21044.4525                                     | 1.02%           | -1.32%          |
|              |               |                          | Absolute Sum of Error %                        | 8.38%           | 9.46%           |

Forecasting was completed through Nov 7<sup>th</sup>, as well as throughout the end of the 2025 year. Predictions were calculated in SAS by coding FORECAST LEAD=8 and evaluated in Excel.

| Future Dates | ARIMA(5,1,1) Predictions | SARIMA(0,1,1)(0,1,1) <sub>52</sub> Predictions |
|--------------|--------------------------|------------------------------------------------|
| Nov 14, 2025 | 20508.6579               | 20562.5146                                     |
| Nov 21, 2025 | 20540.4963               | 20288.3764                                     |
| Nov 28, 2025 | 20546.5386               | 20251.6289                                     |
| Dec 05, 2025 | 20551.1202               | 21266.5324                                     |
| Dec 12, 2025 | 20549.7422               | 20904.9783                                     |
| Dec 19, 2025 | 20551.3998               | 21123.041                                      |
| Dec 26, 2025 | 20555.0793               | 19353.3327                                     |
| Jan 02, 2026 | 20557.7175               | 20146.9142                                     |

Both models produce similar forecasts, keeping errors below 10%, with ARIMA performing slightly better than SARIMA, indicating reasonable predictive ability. Still, choosing a model based on only a month of performance is not ideal for long term use. The ARIMA forecasts show a steady upward trend in Petroleum Products Supplied, while SARIMA introduces more fluctuation over time. Ultimately, model selection should consider not just short-term accuracy but also AIC and SBC values.

## Model Comparison

Both the ARIMA(5,1,1) and SARIMA(0,1,1)(0,1,1)<sub>52</sub> models provide valid representations of weekly U.S. petroleum product demand, but each does have its limitations. The ARIMA model captures short term dynamics well and produced slightly lower forecast error over the evaluation period, but it does not incorporate potential seasonality that may be relevant for annual petroleum consumption. The SARIMA model addresses this by including seasonality, and it achieves better AIC and SBC values, but its short-term forecasts do not clearly outperform ARIMA's accuracy. These weaknesses could be reduced by refining the model structure, testing alternative seasonal periods, or exploring other methods to better include outliers. From a business perspective, both models suggest stable post-2020 demand with gradual recovery toward pre-pandemic levels. For operational planning, the steadier ARIMA forecasts may support short term production decisions, while the SARIMA model may be more informative for identifying possible seasonal shifts over longer timeframes. Considering overall model fit, seasonal structure, and forecast behavior, the SARIMA model is preferred for practical use. Its stronger SBC and AIC indicate a better balance of complexity and explanatory power, and its seasonal patterns will more likely align closer with how petroleum demand varies throughout the year, even if the seasonal signal is small. However, for short term forecasting tasks, the ARIMA model remains a strong secondary option.

## Conclusion

The analysis shows that weekly U.S. petroleum product demand follows predictable seasonal patterns and responds sharply to major economic disruptions yet remains stable over the long term. Both the ARIMA(5,1,1) and SARIMA(0,1,1)(0,1,1)<sub>52</sub> models produced reasonable forecasts, with ARIMA performing slightly better in short-term accuracy and SARIMA offering a stronger overall fit and a clearer representation of yearly seasonality. The forecasted values indicate that total petroleum products supplied will remain near 19 to 20 million barrels per day, suggesting a mature and steady market without significant upward or downward pressure. From a business standpoint, this stability supports consistent refining, stable inventory management, and planning that accounts for seasonal peaks. The forecasts imply that capacity is not changing quickly, and businesses should focus on efficient and sustainable operations in the long term rather than rushing to develop more facilities.